

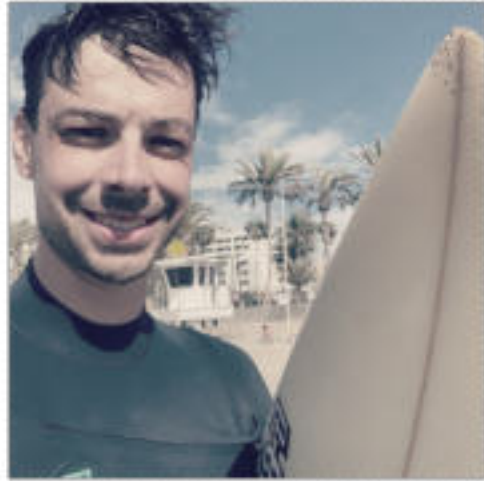
Portfolio

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FLOW-THROUGH LOGISTICS

- seeking simplicity and clarity in architecture
- blending creativity with technical bcg
- constantly exploring: painting, 3d-modelling, simulations, structural engineering



FAVOURITE QUOTE

“Simplicity is the ultimate sophistication.”

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Case study

MANEUVERS FOOTPRINT



90° sight-side alley dock

the truck pulls past the target bay parallel to the dock face, then jackknifes the trailer ~45° and reverses while the trailer swings into the slot between adjacent rigs.

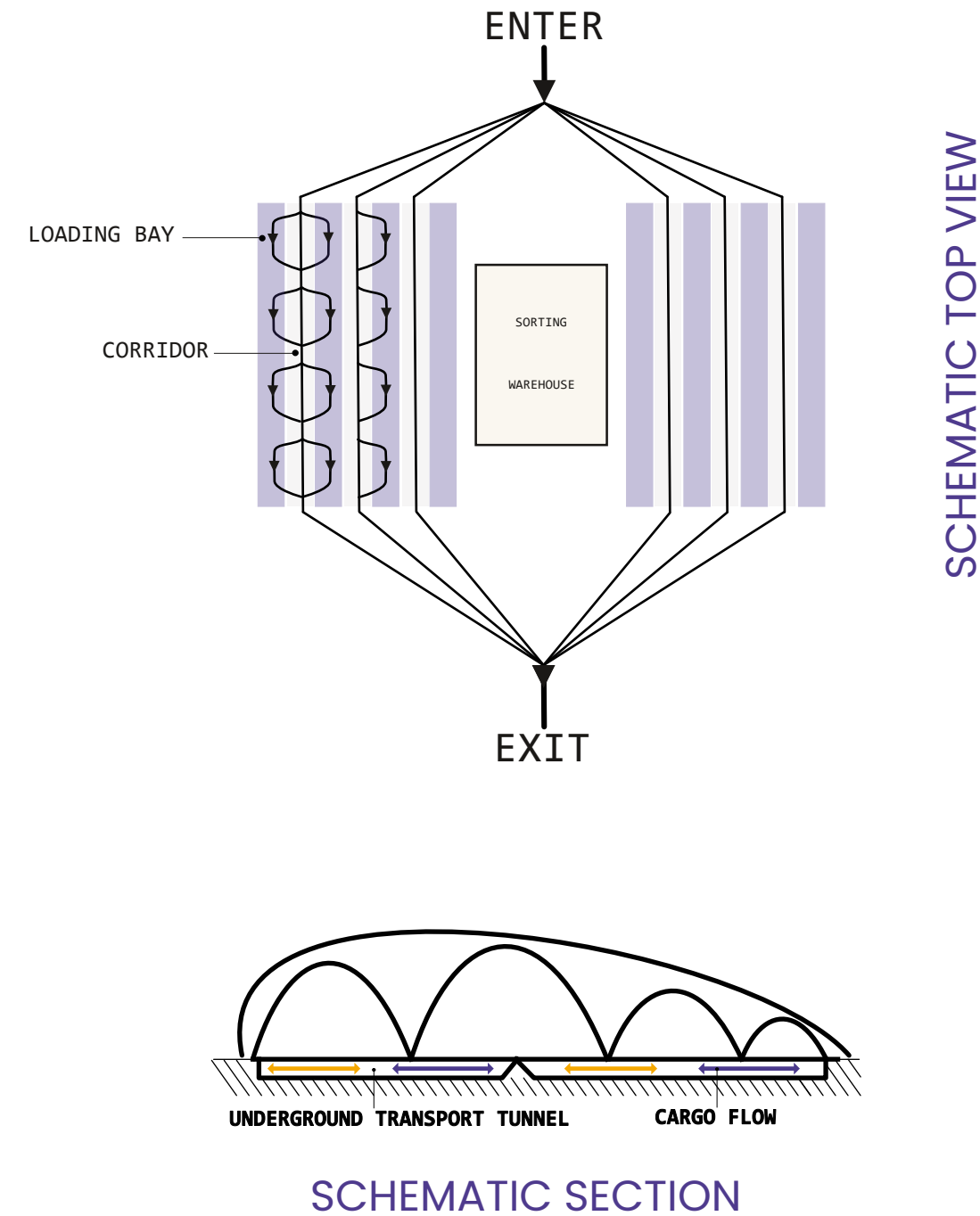
45° approach turn

the driver loops out from the drive lane into the open apron to set up a shallower angle to the bay, then reverses with a smaller jackknife and less swing — trading additional forward maneuvering for a simpler backing motion.

Both demand a substantial maneuvering area in front of the dock: typically 30–40 m of clear apron depth for a standard 16.5 m articulated truck, plus lateral clearance for the pull-past and the trailer's swept path.

Without this footprint, neither maneuver can be executed safely without multiple pull-up corrections.

FLOW-THROUGH



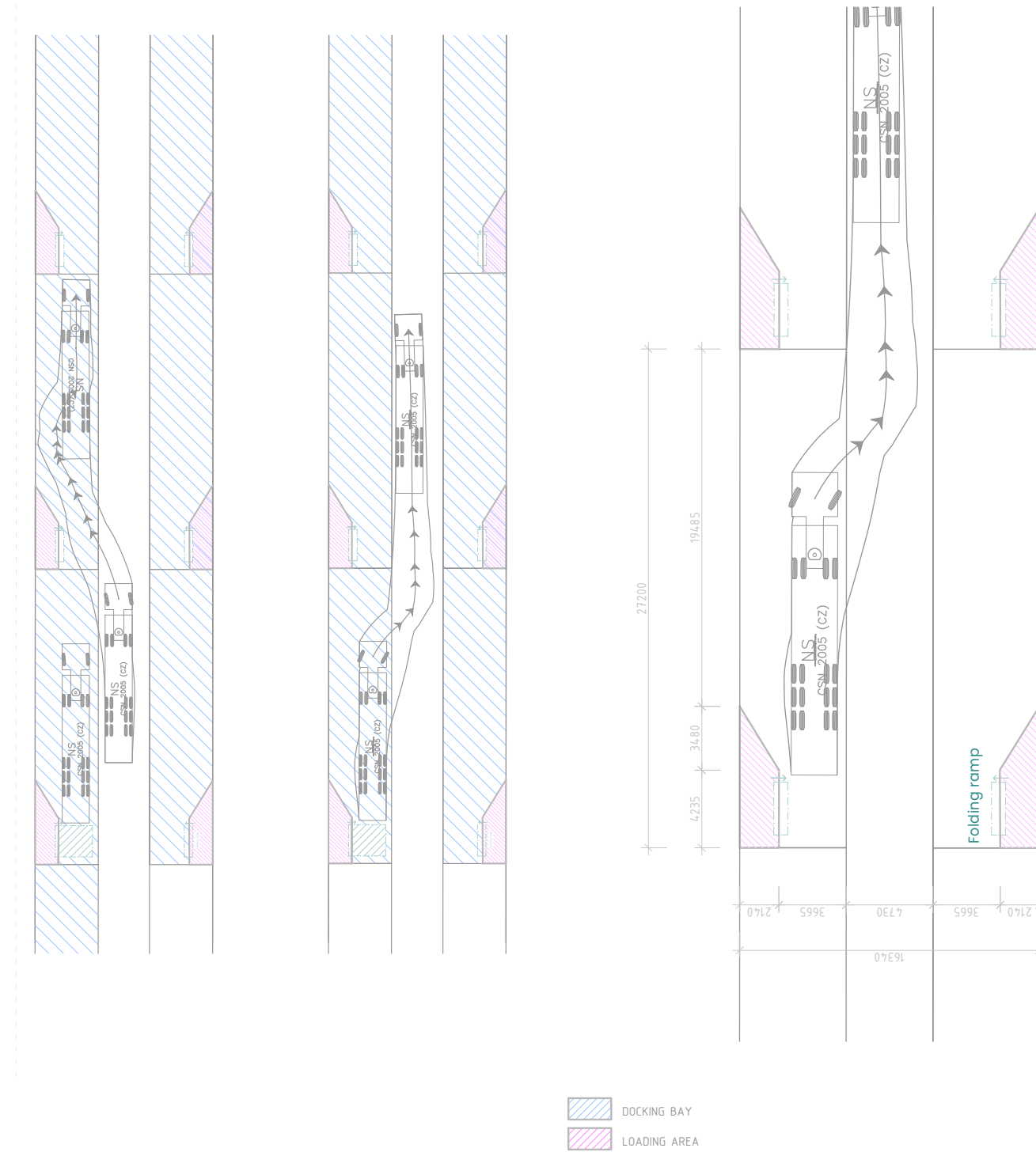
This concept replaces multi-point maneuvers with a flow-through configuration:

1. **truck enters a corridor**
2. **switch lanes to align with a loading bay**
3. **reverse shortly to dock**
4. **exit forward**

Faster dock cycles allow the same throughput to be served by fewer docks, reducing total paved surface.

Cargo continuity inside the facility is sustained by shallow underground transport tunnels running beneath the loading bays, decoupling cargo flow from truck movement above.

GEOMETRIC VALIDATION

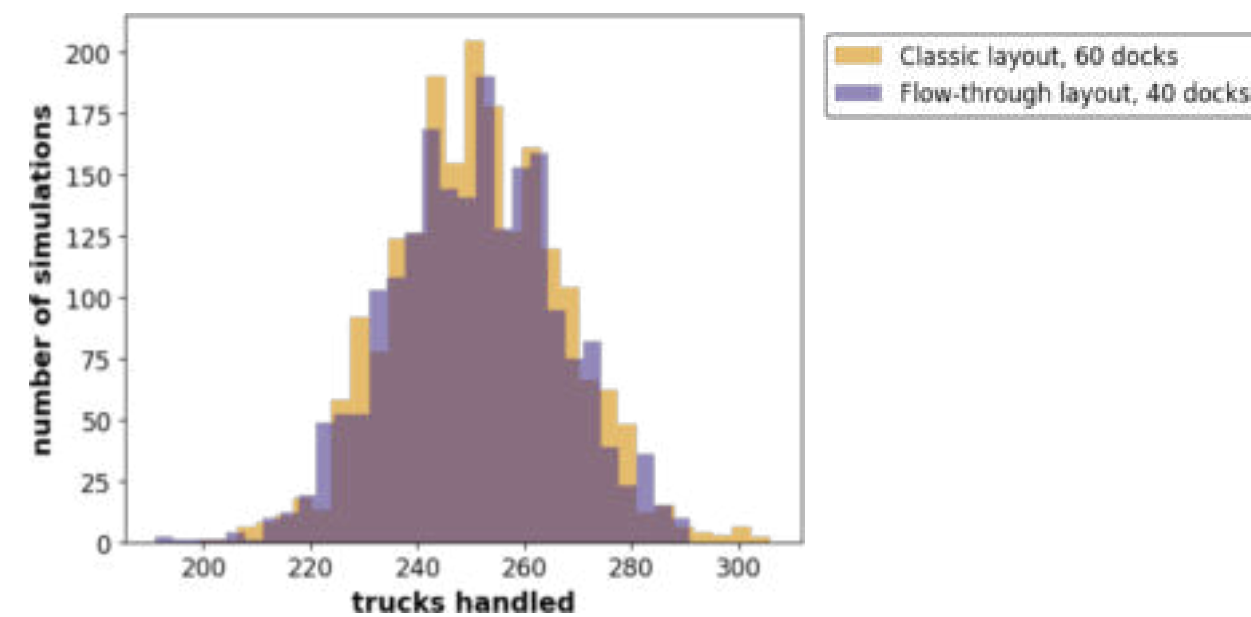
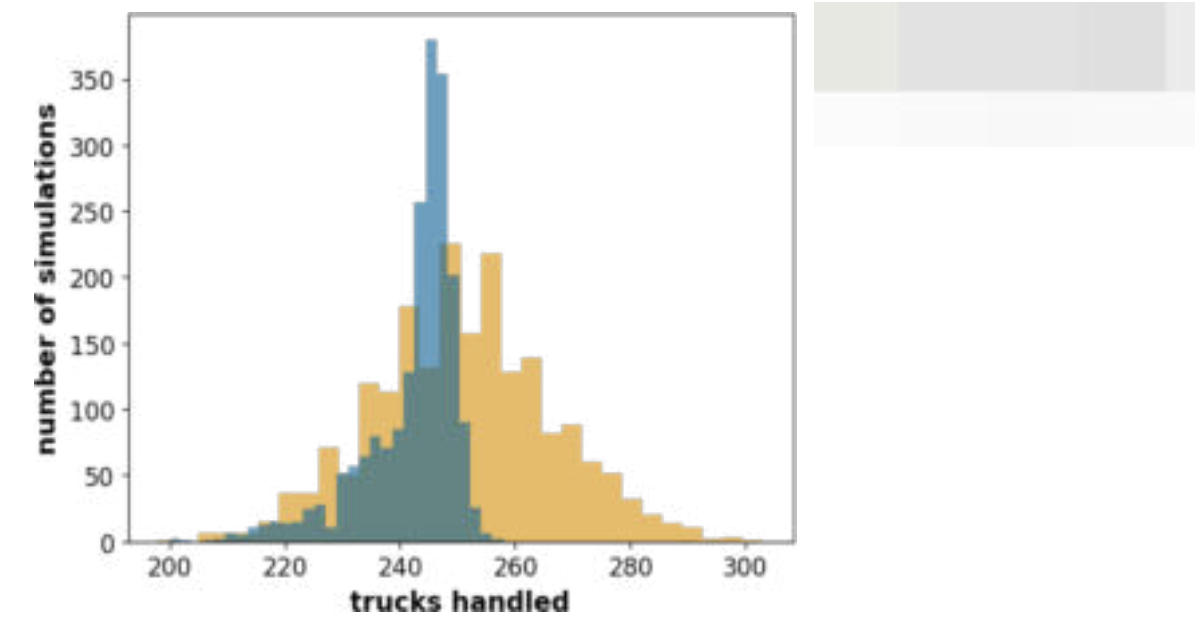


Swept path analysis was conducted to validate and establish the initial dimensions of each docking bay and loading area, ensuring conformity with standard truck dimensions and the required maneuvering envelope.

AutoTURN software was employed to generate the specific swept paths, and based on these results, both the width and length of the loading bays were subsequently increased.

To further capitalize on the proposed concept, a **folding ramp** has been introduced. This solution permits a standard truck to be loaded from the rear while maintaining the minimum feasible bay length.

OPERATIONAL VALIDATION

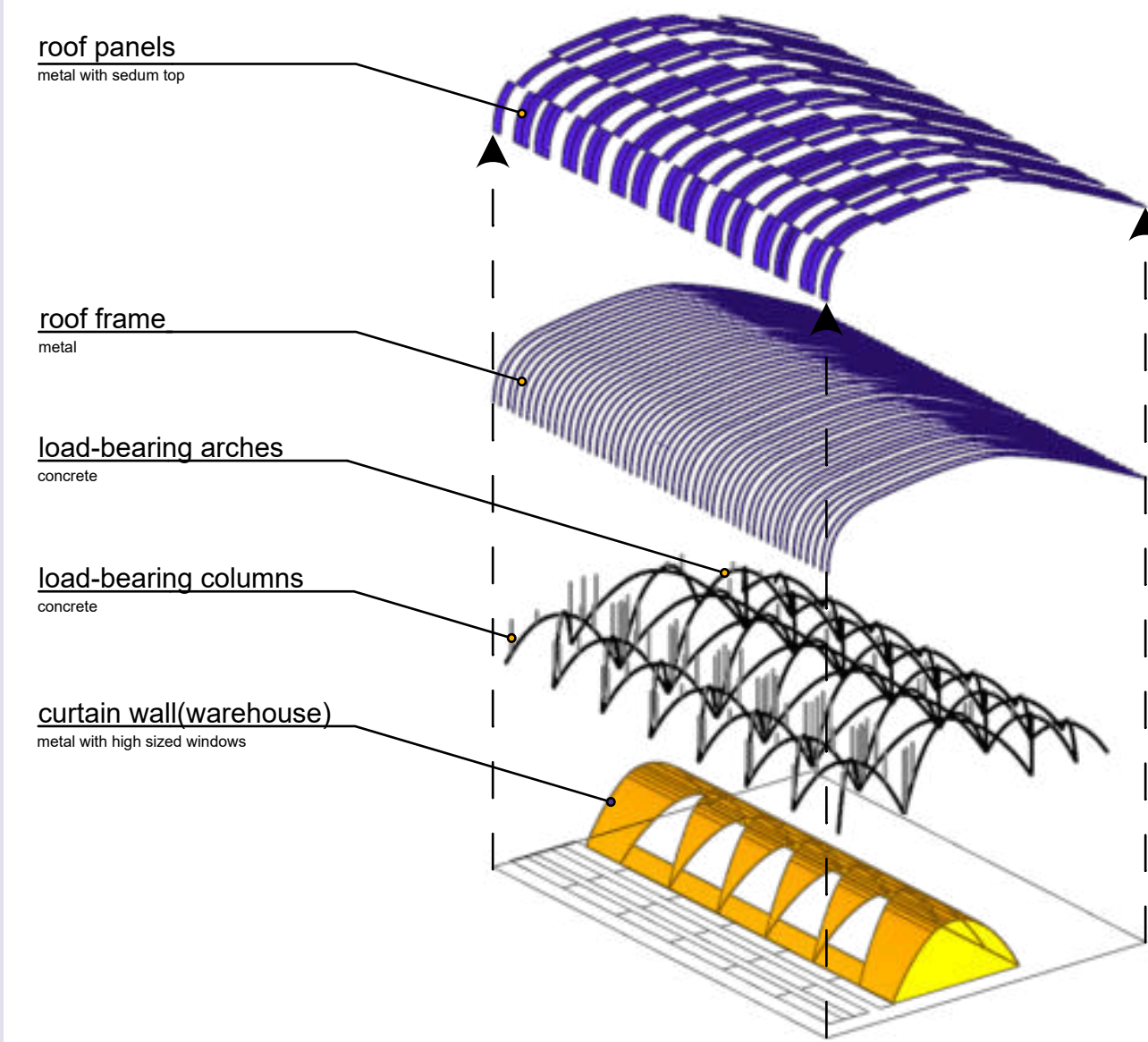


A discrete-event **simulation** was developed to test whether the flow-through configuration can match the throughput of a conventional layout while operating with fewer docks. For all simulations average arrival rate was considered 250 trucks/hour. In total result 2000 runs of simulation for each configuration was used.

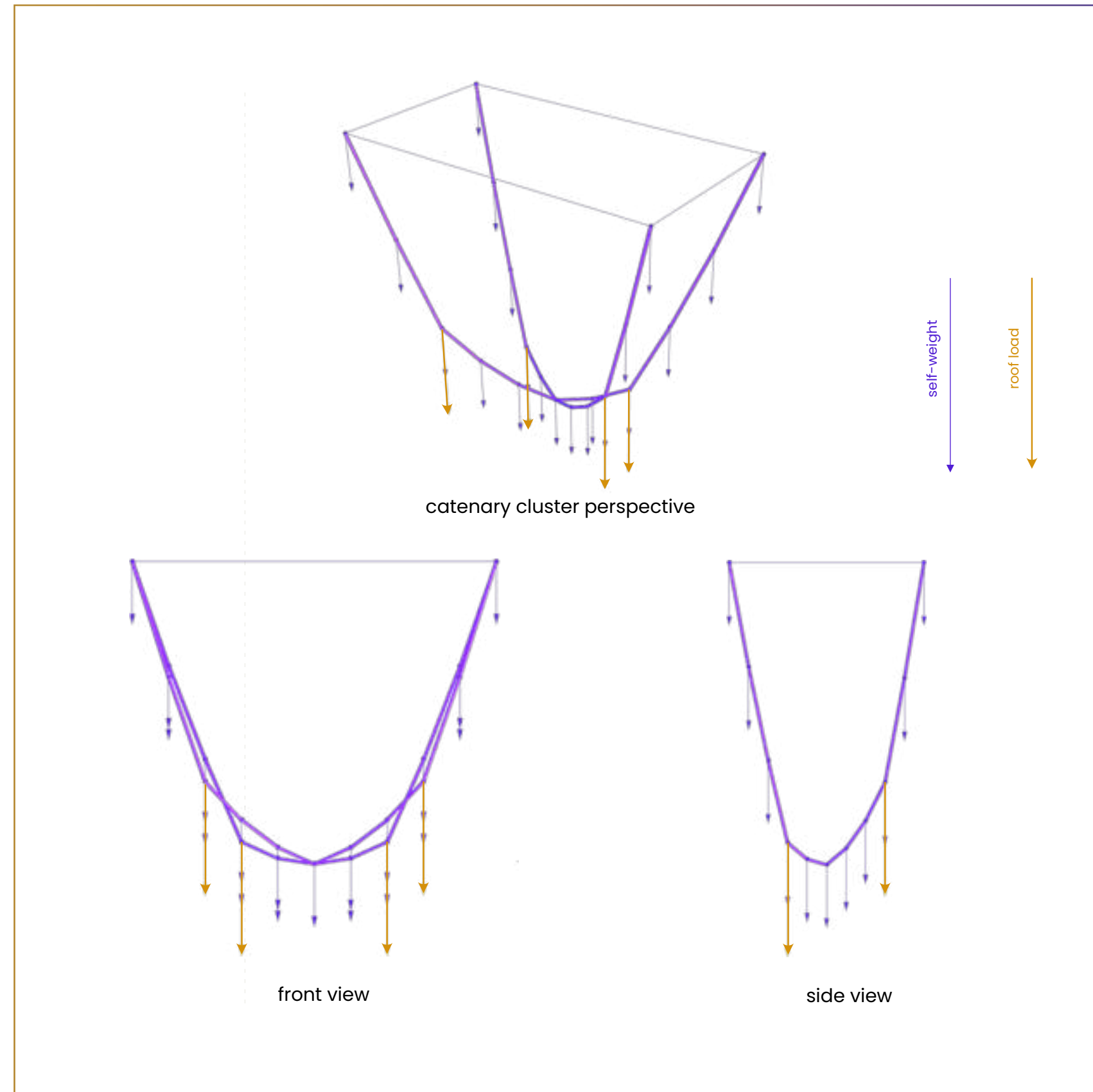
The first plot compares the classic layout at 60 docks (yellow) against the same classic layout reduced to 40 docks (blue). The reduced case collapses into a narrow distribution peaking around 245 trucks per day, well below the 60-dock case and with markedly lower variance — the system has **bottlenecked**. This establishes that dock count is genuinely consequential in the conventional layout: docks cannot simply be removed without a throughput penalty.

The second plot compares the same classic layout at 60 docks (yellow) against the flow-through layout at 40 docks (purple). Here the two distributions overlap almost completely, both centered near 250 trucks per day with comparable spread. **With the same dock reduction that bottlenecked the classic layout, the flow-through configuration recovers full throughput.**

CONSTRUCTION



Top view



A catenary-based **form-finding technique** was used to generate the load-bearing arches. A hanging chain simulated under self-weight and tributary roof loads produces a funicular curve — once inverted, the geometry carries loads in pure **axial compression with zero bending moment**.

Using the **Kangaroo** plugin for Rhino, two types of loads were simulated: the first is **self-weight** distributed evenly along the curve; the second is the **load introduced by the roof panels**. The position of these load points was dictated solely by the location of the roof panels, propagating their specific spatial arrangement.

The arrangement shown in the portfolio follows an aesthetic standpoint, but it could be optimized for a specific purpose.



main loading bays perspective

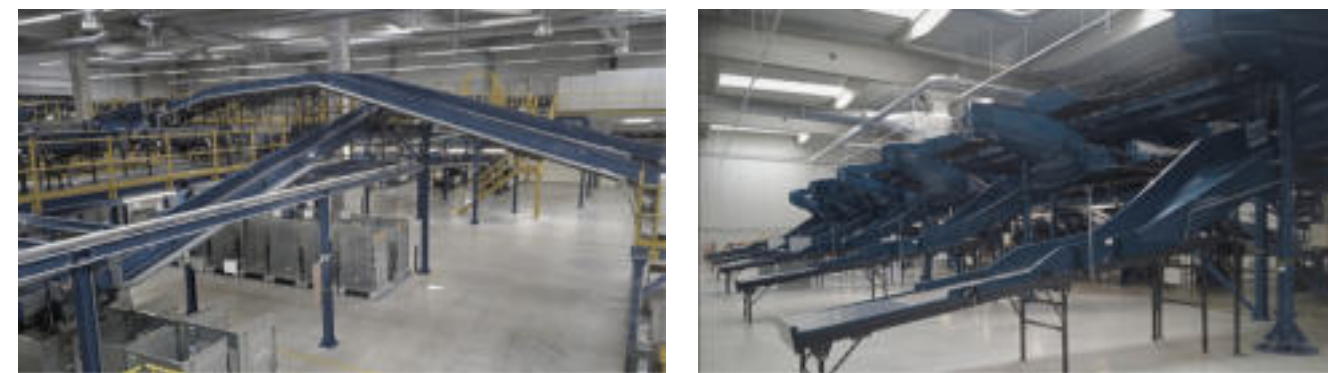
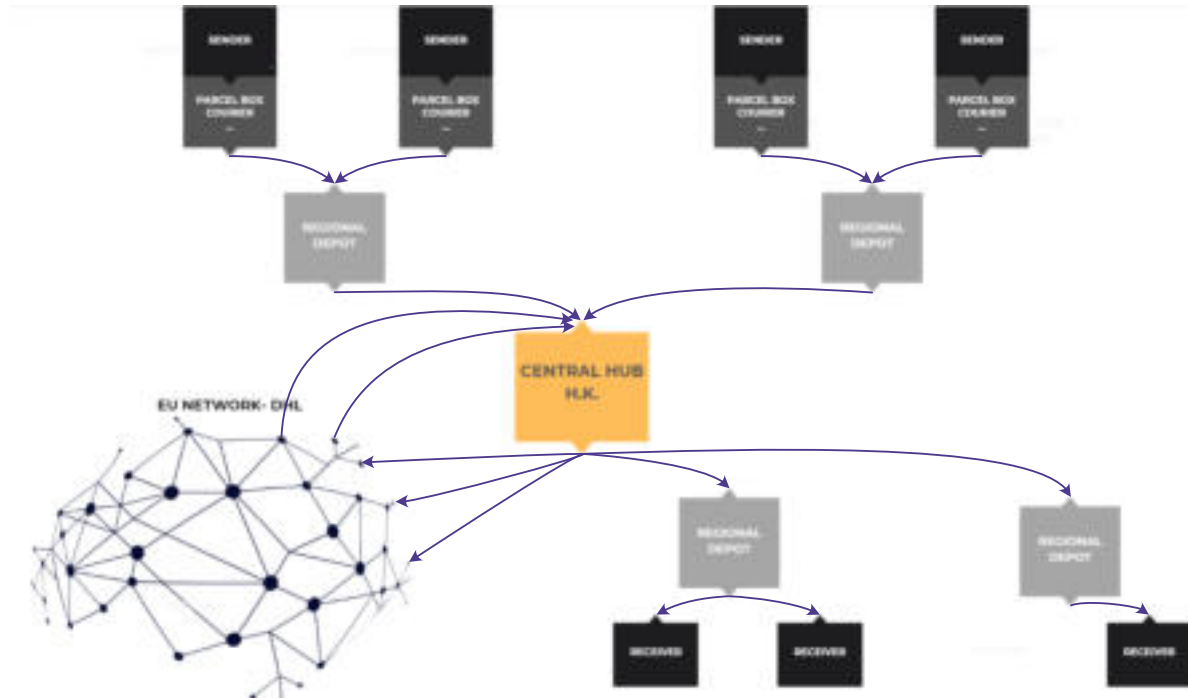
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roof edge detail

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PPL CASE STUDY



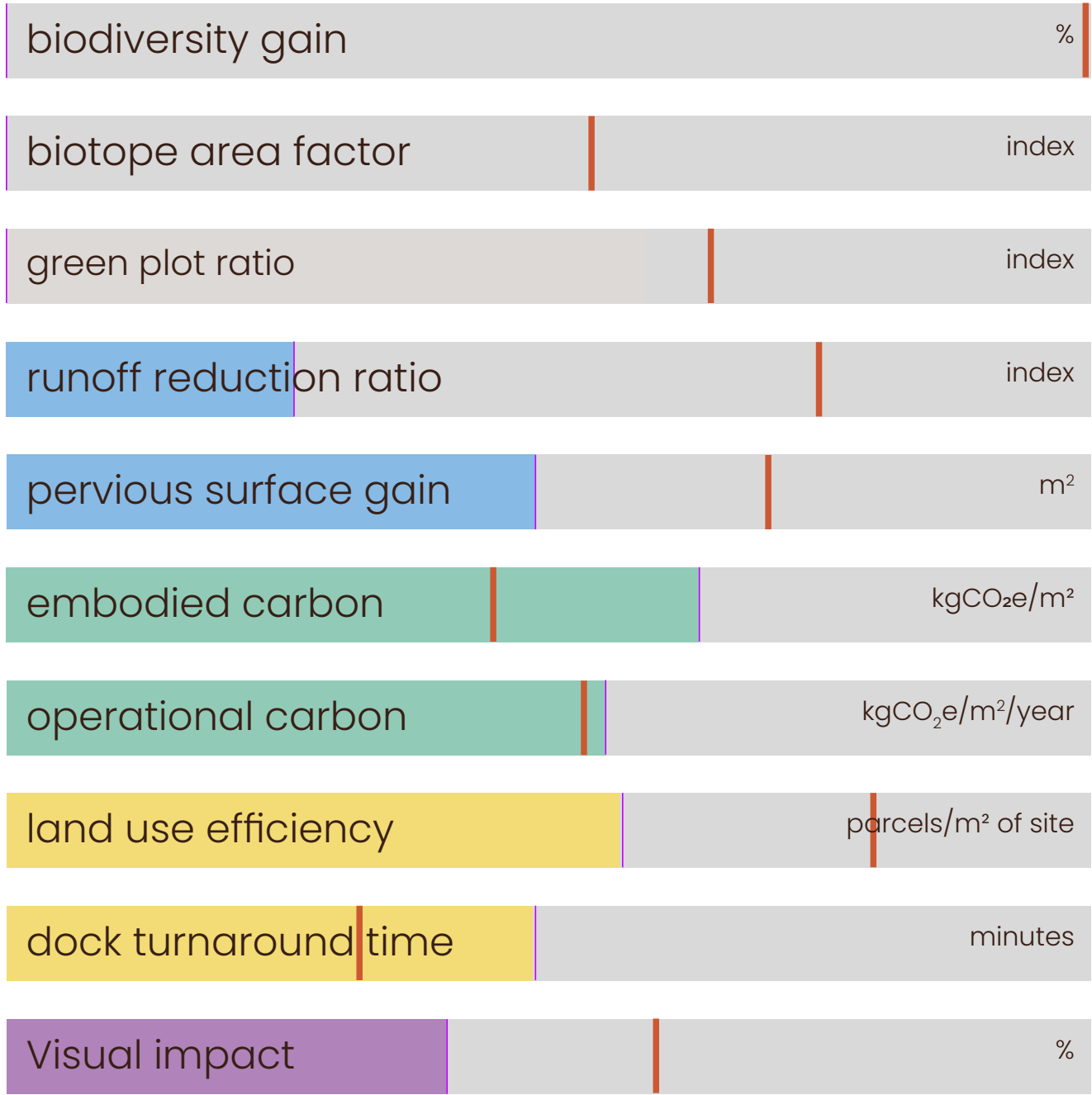
PPL **hub**, a fully automated e-commerce sorting centre. The facility operates with distinct inbound and outbound traffic peaks – late evening and night for inbound, late night and early morning for outbound – with a dock ratio of 20:50 in favor of outbound. Therefore it is assumed a total of 70 docks are present at facility. The hub has critical as PPL is subsidiary of DHL. Therefore it serves as **gateway to EU DHL network**.

Integration of conveyor belts into the docking bays is considered feasible given the facility's existing automation infrastructure. Belt routing via underground tunnels or overhead systems could be both viable alternatives.



Applying a flow-through logistics concept at PPL Hradec Králové enabled a **20-dock reduction** without compromising performance – the simulation confirmed an unchanged throughput when 300 trucks/hour. This translates directly into a **30% decrease in paved (impervious) surface area**

KPIs



Description

- biodiversity gain** - quantifies the percentage increase in "biodiversity units" achieved through habitat creation/restoration versus the site baseline.
- biotope area factor** - measures how much of the plot functions ecologically, based on weighted surface categories.
- green plot ratio** - considers the total leaf are of plants in relation to site area. Unlike purely surface-based metrics, it values the density and volume of planting through LAI.
- runoof reduction ration** - percentage of stormwater retained on-site compared to the pre-development reference state.
- pervious surface gain** -net increase in permeable surface area compared to the pre-development reference state.
- embodied carbon** - the CO2 emmited during extraction, manufacture, transport and assembly of building materials.
- operational carbon** - estimated reduction in carbon emissions generated during the building's day-to-day operations compared to a conventional logistics facility of equivalent capacity.
- land use efficiency** - economic output generated per square metre of occupied site area, measured as parcels processed per day per m² of gross floor area.
- dock turnaround time** - average time elapsed between a truck arriving at the facility gate and departing the site, encompassing all operational phases.
- visual impact** - percentage of the facility's built perimeter screened from surrounding public viewpoints by vegetation, green buffers, or landscape features, compared to the pre-development.reference state.